SHORT PROJECT REPORT

Weather Data Analyzer & Temperature Prediction System

Introduction to Open Source Software · Sejong University · June 2026
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DATASET	LANGUAGE	BEST R^2 SCORE
Seoul Weather	**Python 3.10+**	**76.98%**

1 Project Purpose & Description

Understanding temperature fluctuations and weather trends is critical for meteorological forecasting, agricultural planning, and energy grid load predictions. This desktop application provides an end-to-end framework to load weather datasets, automatically clean and impute missing parameters, visualize trends, and fit predictive machine learning models.

Why is this useful? Meteorologists, local municipalities, and energy providers can leverage historical statistics and linear predictions to anticipate climate anomalies, evaluate environmental coefficients, and optimize grid scheduling to prevent peak overload.

Auto Column Mapping
Smart regex detects date, temp, wind, precip, and conditions columns dynamically.

Missing Data Cleaning
Cleans voids by imputing numerical values with means and categories with modes.

7 Visualizations
Saves monthly averages, histograms, trendlines, and correlation heatmaps to disk.

Linear Regressor Engine
Splits dataset features 80/20 to fit scikit-learn models and outputs accuracy.

Interactive Prediction Form
Allows live predictions by manually entering weather features in the GUI.

Academic MIT License
Fully open source, free to redistribute, modify, and integrate for coursework.

2 Tools & Libraries Used

pandas
Data manipulation & CSV parsing

numpy
Vectorized math & array logic

scikit-learn
Predictive modeling engine

matplotlib
Core plot rendering backend

seaborn
Statistical heatmap visuals

tkinter
Desktop GUI window interface

All dependencies are open source and installable in a single command: pip install -r requirements.txt

3 Descriptive Statistics Summary

Weather Metric	Average / Total Value	Maximum Recorded	Minimum Recorded
Temperature (°C)	13.69 °C	31.10 °C	-13.70 °C
Humidity (%)	65.28 %	99.00 %	22.00 %
Rainfall (mm)	4.35 mm (Avg)	268.62 mm	3178.11 mm (Total)
Wind Speed (m/s)	14.37 m/s	33.00 m/s	N/A

Weather Conditions: Most Common Condition is **Partially cloudy**, and the Least Common Condition is **Snow, Overcast**.

4 Project Workflow

1. **Load Dataset** — Loads local weather CSV files dynamically via native Tkinter file dialogs.
2. **Preprocess & Clean** — Sorts data chronologically, parses dates, and imputes missing fields with column means/modes.
3. **Descriptive Statistics** — Calculates aggregated average, maximum, and minimum parameters for weather variables.
4. **Save Visualizations** — Saves 7 styled PNG plots illustrating historical trends, densities, and correlations.
5. **Train Regressor** — Splits the processed numerical attributes 80/20 and fits a linear regressor using scikit-learn.
6. **Predict Temperatures** — Accepts live manual input variables in the GUI form and runs prediction instantly.

5 Graphical Data Analysis

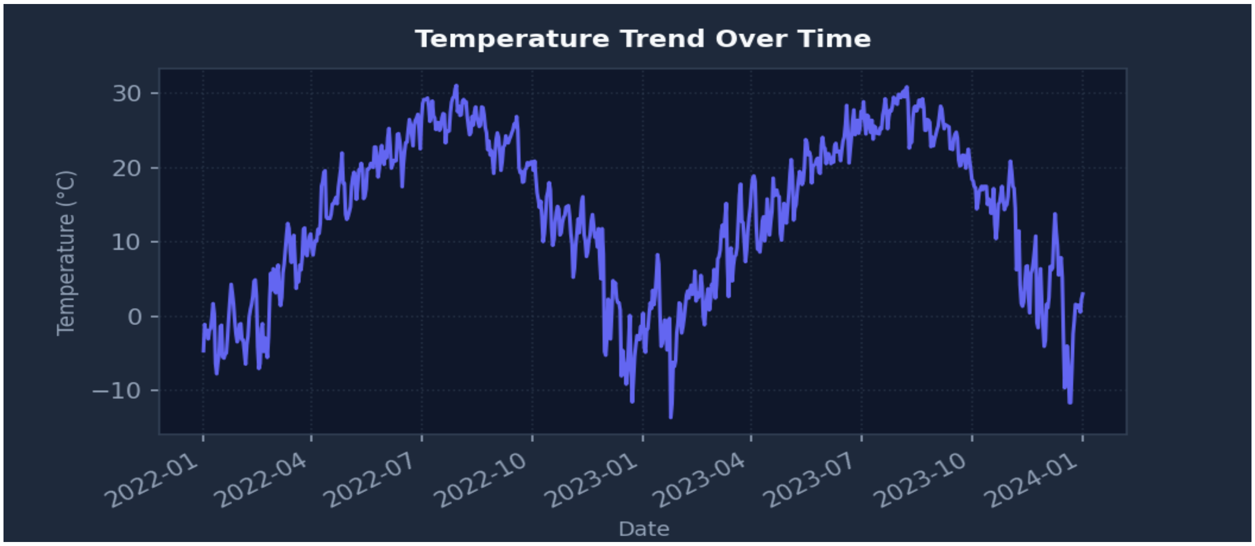

Figure 1: Chronological Temperature Trend Over Time

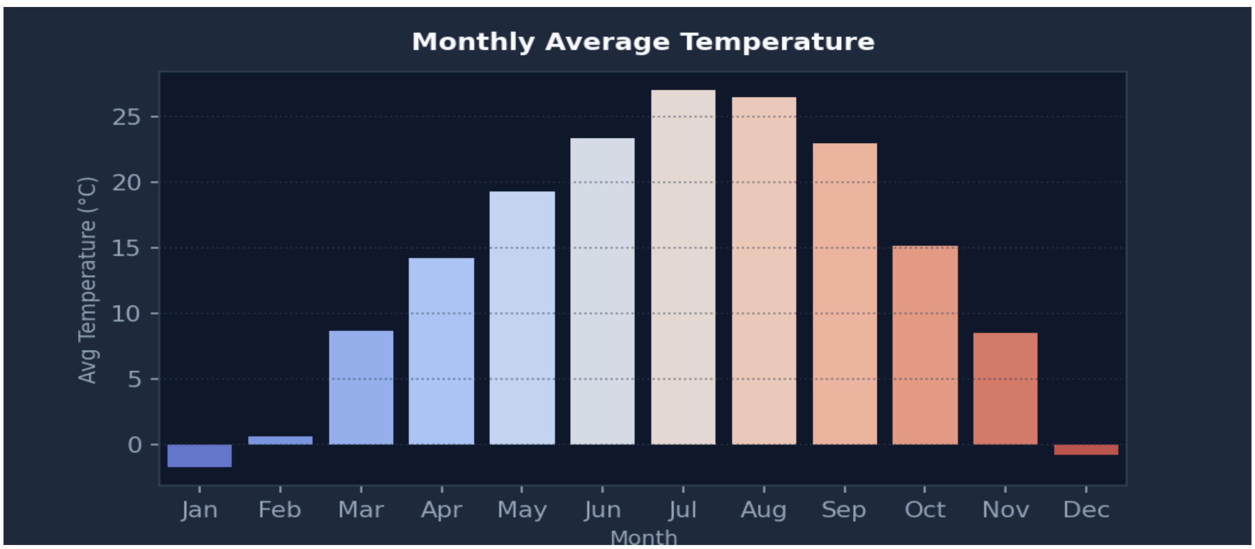

Figure 2: Monthly Average Temperature Comparison

5 Graphical Data Analysis (Cont.)

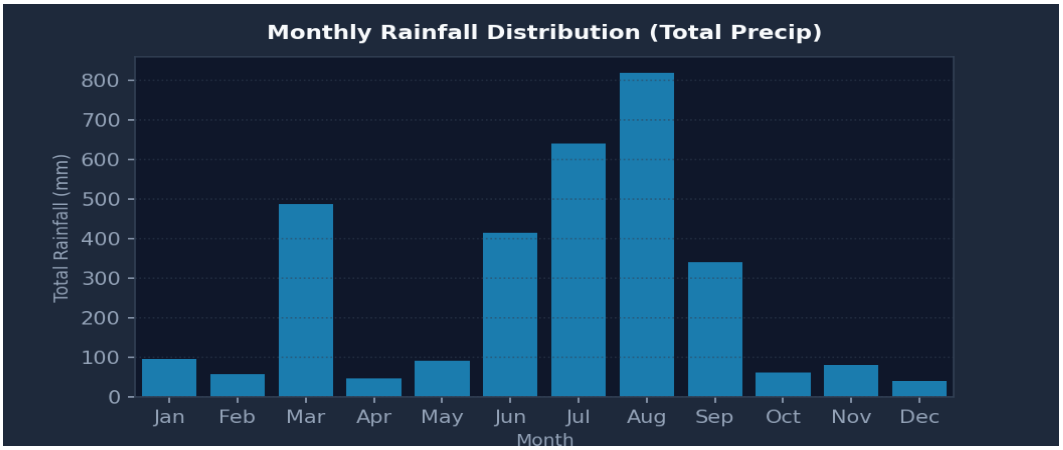

Figure 3: Monthly Rainfall Distribution (Total Precip)

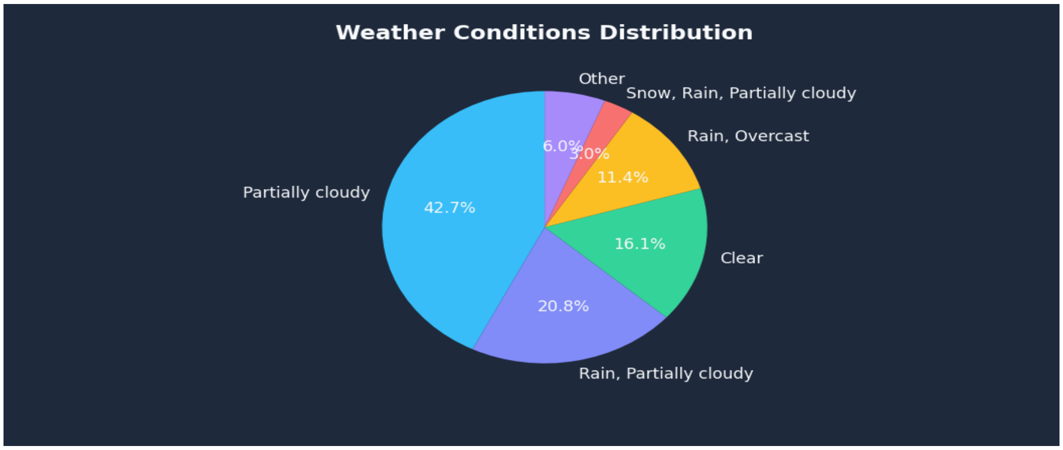

Figure 4: Weather Conditions Frequency Distribution

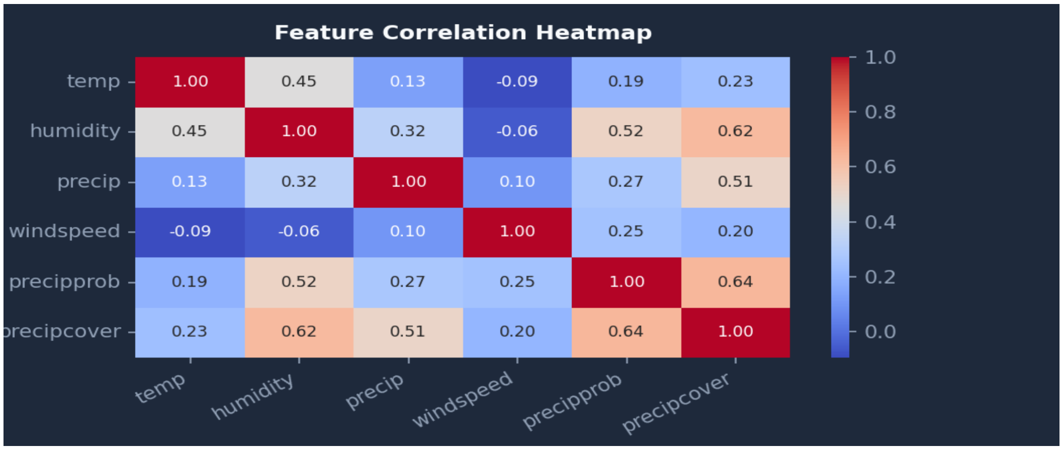

Figure 5: Feature Correlation Heatmap Matrix

6 Machine Learning Results & Performance

A **Linear Regression** model was trained to predict the target temperature variable **temp**. The model was successfully fitted on **584 samples** (80%) and evaluated against **147 unseen test samples** (20%).

Evaluation Metric	Value	Interpretation / Description
Mean Absolute Error (MAE)	4.1071 °C	Average magnitude of errors in predictions.
Mean Squared Error (MSE)	27.5747	Average squared difference between predicted and actual.
Root Mean Squared Error (RMSE)	5.2512 °C	Standard deviation of prediction residuals.
R^2 Score (R-squared) ■ Best	0.7698	Proportion of target variance explained by the model.

Model Coefficients Summary:

Predictor Feature	Coefficient	Effect on Temperature
precip	0.0249	Positive relationship (Temp rises)
precipprob	-0.0064	Negative relationship (Temp drops)
precipcover	0.0584	Positive relationship (Temp rises)
snow	-0.3882	Negative relationship (Temp drops)
snowdepth	-1.9296	Negative relationship (Temp drops)
windgust	-0.0850	Negative relationship (Temp drops)
windspeed	-0.2832	Negative relationship (Temp drops)
winddir	-0.0090	Negative relationship (Temp drops)
sealevelpressure	-0.7680	Negative relationship (Temp drops)
cloudcover	0.0496	Positive relationship (Temp rises)
visibility	0.1222	Positive relationship (Temp rises)
solarradiation	0.0473	Positive relationship (Temp rises)
solarenergy	-0.1196	Negative relationship (Temp drops)
severerisk	0.0640	Positive relationship (Temp rises)
moonphase	-1.0229	Negative relationship (Temp drops)
[Intercept Constant]	790.6059	Base temperature value

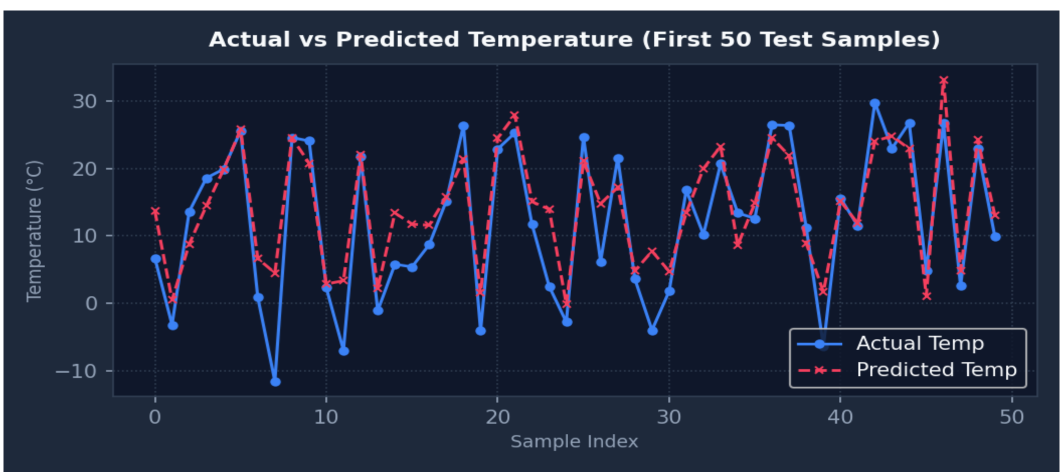

Figure 6: Actual vs Predicted Temperature on Test Set

7 Why README.md and requirements.txt Matter

README.MD

The README acts as the entry portal for any software repository. It answers essential questions for developers and users alike: what does the project do, how do you configure it, and how do you execute it? On platforms like GitHub, the README renders dynamically at the root page, serving as the core reference before anyone contributes to or deploys the software system.

REQUIREMENTS.TXT

This file catalogs the required third-party Python packages and strict version constraints. Executing pip install -r requirements.txt replicates the exact runtime environment on any machine in seconds. This prevents library collisions, fixes reproducibility issues across different operating systems, and ensures seamless academic evaluation.